MIL-G-37331A
8 December 1976
SUPERSEDING
MIL-G-37331 (DSA-DM)
3 October 1975

MILITARY SPECIFICATION

GLUCOSE TEST SOLUTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Glucose Test Solution, 100 Gram, 10 to 12 fl oz (296 to 355 ml) (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals. (See 6.3)

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, Pa. 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and
Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042).

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a pleasantly flavored, carbonated solution containing 100 grams of Dextrose in not less than 10 fl. oz. (296 ml) and not more than 12 fl. oz. (355 ml) of vehicle.

3.2 Additives. The coloring, flavoring, and preservative(s) used in the solution shall be in the amounts approved by the Federal Food and Drug Administration.

3.3 Finished product (solution).

3.3.1 Assays.

3.3.1.1 Glucose (Dextrose). The solution shall assay to contain not less than 92.5 percent and not more than 107.5 percent of the required amount of dextrose, when determined as specified in 4.5.1.

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3.3.2 Identification.

3.3.2.1 Dextrose. Shall be identified in accordance with the USP procedure.

3.3.2.2 Carbon Dioxide. Shall be identified in accordance with the USP procedure in the monograph for Carbon Dioxide.

3.3.3 Carbon Dioxide content. Shall contain not less than 1.0 volume when determined as specified in 4.5.2.

3.3.4 pH. Shall not exceed 4.00 when determined by the USP potentiometric method at 25°C.

3.3.5 Microbiological tests. Shall contain no more than 100 colonies per ml. Coliform organisms shall not be present. The yeast and mold counts shall each not exceed a maximum of 10 organism per ml. Testing shall be in accordance with 4.5.3.

3.4 Flavor and platability. The finished solution shall be palatable and pleasant to the tast with no unpleasant after-taste. Not later than the time specified for opeining of bids or receipt of proposals, the offeror shall submit to the contracting officer six (6) individually packaged samples of the finished solution representative of the product which the offeror proposes to furnish. Two (2) samples will be subjected to panel testing for a determination of palatability (see 4.6 - Palatability test). The remaining samples will be used by cognizant Government inspection and quality assurance activities for determining compliance of supplies furnished hereunder with the palatability requirement. Approval as to palatability of any sample submitted by the offeror will not constitute approval of the sample as to any other requirement of this specification. The requirement for submission of samples for use in determining compliance with the palatability requirement may be waived, provided the offeror states, in his bid or proposal, that the product he proposes to furnish is the same product he has offered to the purchasing activity on a previous procurement and the contracting officer determined that such product was previously procured and/or tested by the purchasing activity and found to comply with the palatability requirement.

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3.5 Ingredients.

3.5.1 Dextrose and carbon dioxide. The dextrose, and carbon dioxide used in the manufacture of the solution shall be in accordance with the tests, standards, and requirements of the USP including any supplements or revisions thereto.

3.6 Other ingredients. All other ingredients entering into the finished solution shall be of USP or NF quality or, if not included in either of these compendia, the ingredients shall be suitable for use in this preparation.

3.7 Expiration dating period. The expiration dating period shall be not less than 36 months.

3.7.1 Delivery. Not more than 3 months of the expiration dating period shall have elapsed at the time of delivery to the Government.

3.8 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of solution produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

Table I. Sampling

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Filled immediate container

4.3.1 For end item testing. Noncompliance with any of the following is cause for rejection. The acceptance number for each test shall be zero (0).

Table II. Criteria for End item testing

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Assays	3.3.1	S-2	5) Composite
Identification	3.3.2	S-2	5)
Carbon Dioxide Content	3.3.3	S-2	3
pH	3.3.4	S-2	5) Composite
Microbiological Tests	3.3.5	S-2	5)

1/Where possible, the sample shall be used for two or more tests.

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4.4 Examination. The finished solution shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

Table III. Classification of defects

Categories	Defects 1/
Major	
101	Solution not free from foreign matter.
102	Solution has crystal formation.
103	Solution not homogeneous.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks and tears.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Tests.

4.5 Assays.

4.5.1 Dextrose. Shall be determined by an appropriate, accurate, and reproducible method. The bidder or offeror shall submit with its bid or proposal the detailed proposed method. Assay method shall be validated by submission of an adequate number of test results including date of assay, lot number, mg of dextrose per ml found, percentage of required amount and statistical data including as a minimum the Standard Deviation and Relative Standard Deviation. If the method is approved and accepted by the Government, such method shall be employed in the testing of the solution, and need not be resubmitted, provided the bidder or offeror certifies that there has been no change in formulation or method of manufacture.

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4.5.2 Carbon dioxide content. Volume of gas shall be determined by a Zahm gauge, or equivalent instrument capable of piercing a metal bottle cap which shall furnish an accurate reading in pounds per square inch of gas pressure. Volume of carbon dioxide in the immediate container is determined by reference to a suitable table correlating pressure, temperature, and gas volume.

4.5.3 Microbiological tests. Shall be determined by appropriate, accurate, and reproducible methods. Separate culture media shall be used for the bacterial count, and for the yeast and mold count.

4.6 Palatability test. A taste panel consisting of 10 members will be used to determine acceptability of samples. Samples will be prepared for testing (samples will be tested undiluted) coded, and served to the panel members under controlled serving conditions, e.g., all samples will be the same amount (10 ml), and served at the same temperature; each panel member will receive an equal number of samples; the order of serving will be varied among panel members; an interval of at least five (5) minutes will elapse between successive samples and panel members will rinse their mouths with water (room temperature) after each sample; panel members will test without interference either from each other or from outsiders. The product offered shall be rated equal to or better than the Palatability Standard 1/, when determined by the taste panel, using the following 9-point hedonic rating scale. The average rating of the sample shall be equal to or greater than the average rating of the standard, similarly prepared and tested.

1	2	3	4	5	6	7	8	9
Dislike extremely	Dislike very much	Dislike moderately	Dislike slightly	Neither like nor dislike	Like slightly	Like moderately	Like very much	Like extremely

1/ Glucose Test Solution Palatability Standard is available, upon separate request to the contracting officer, Defense Personnel Support Center.

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4.7 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.8 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 and as specified herein.

5.2 Packaging. Packaging shall be Level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 10 to 12 fl. oz. (296 to 355 ml) of solution as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, a unit package may be furnished. If a unit package is furnished, each unit shall be packaged as specified in 5.2.4 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve (12) units of issue shall be packaged as specified in 5.2.8 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.3 Packing. Packing shall be level A, B or C as specified (See 6.1).

5.3.1 Level A. Four intermediate packages (48 units) shall be packed in an exterior container as specified in 5.4.4.4 of PPP-C-186.

5.3.2 Level B. Four intermediate packages (48units) shall be packed in an exterior container as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the National Stock Number.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

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5.4.2 Intermediate package. Each intermediate package shall be marked in accordance with 5.5.3 of PPP-C-186. Type I Shelf life marking as specified in MIL-STD-129 shall also be shown. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F.)"

5.4.3 Exterior container. Exterior container shall be marked in accordance with 5.5.4 of PPP-C-186. Type I Shelf life marking as specified in MIL-STD-129 shall also be shown. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F.)"

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-00-132-0373	GLUCOSE TEST SOLUTION, 100 Gram, * fl oz (* ml)

* insert volume in fluid ounces and milliliters of items being supplied.

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1721

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-G-37331A GLUCOSE TEST SOLUTION

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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